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1 CREATE TABLE EmployeeDetails (  
2     EmpId INT PRIMARY KEY,  
3     FullName VARCHAR(100),  
4     ManagerId INT,  
5     DateOfJoining DATE,  
6     City VARCHAR(100)  
7 );  
8  
9 CREATE TABLE EmployeeSalary (  
10     EmpId INT,  
11     Project VARCHAR(50),  
12     Salary DECIMAL(10,2),  
13     Variable DECIMAL(10,2),  
14     FOREIGN KEY (EmpId) REFERENCES EmployeeDetails(EmpId)  
15 );  
16  
17 INSERT INTO EmployeeDetails  
18 (EmpId, FullName, ManagerId, DateOfJoining, City)  
19 VALUES  
20 (101, 'Alice Johnson', 321, '2022-05-15', 'New York'),  
21 (102, 'Bob Smith', 876, '2020-03-12', 'Los Angeles'),  
22 (103, 'Charlie Brown', 986, '2021-08-23', 'Chicago'),  
23 (104, 'David Williams', 876, '2019-11-05', 'Houston'),  
24 (105, 'Eve Davis', 321, '2023-01-07', 'Phoenix'),  
25 (106, 'Frank Miller', 986, '2018-12-19', 'Philadelphia'),  
26 (107, 'Grace Wilson', 876, '2022-03-28', 'San Antonio'),  
27 (108, 'Hank Moore', 321, '2021-09-14', 'San Diego'),  
28 (109, 'Ivy Taylor', 986, '2020-02-11', 'Dallas'),  
29 (110, 'Jack Anderson', 876, '2022-11-30', 'San Jose'),  
30 (111, 'Karen Thomas', 321, '2021-07-16', 'Austin'),  
31 (112, 'Liam Jackson', 986, '2023-04-21', 'Fort Worth'),  
32 (113, 'Mia White', 876, '2019-06-03', 'Columbus'),  
33 (114, 'Noah Harris', 321, '2020-12-10', 'Charlotte'),  
34 (115, 'Olivia Martin', 986, '2021-10-25', 'San Francisco'),  
35 (116, 'Paul Garcia', 876, '2023-07-18', 'Indianapolis'),  
36 (117, 'Quinn Martinez', 321, '2022-09-07', 'Seattle'),  
37 (118, 'Rachel Rodriguez', 986, '2020-01-15', 'Denver'),  
38 (119, 'Steve Clark', 876, '2021-03-19', 'Washington'),  
39 (120, 'Tina Lewis', 321, '2019-08-31', 'Boston');  
40  
41 INSERT INTO EmployeeSalary  
42 (EmpId, Project, Salary, Variable)  
43 VALUES  
44 (101, 'P1', 7500, 500),  
45 (102, 'P2', 9200, 700),  
46 (103, 'P1', 6700, 600),  
47 (104, 'P3', 8300, 900),  
48 (105, 'P2', 7800, 800),  
49 (106, 'P3', 9100, 1000),
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50 (107, 'P1', 6200, 400),
51 (108, 'P2', 8800, 750),
52 (109, 'P3', 9500, 1100),
53 (110, 'P1', 7200, 650),
54 (111, 'P2', 8700, 850),
55 (112, 'P3', 9300, 1200),
56 (113, 'P1', 7900, 550),
57 (114, 'P2', 6800, 450),
58 (115, 'P3', 8400, 900),
59 (116, 'P1', 7600, 500),
60 (117, 'P2', 8900, 1000),
61 (118, 'P3', 9200, 1100),
62 (119, 'P1', 8100, 600),
63 (120, 'P2', 8300, 750);
64
65 select * from EmployeeDetails;
66 select * from EmployeeSalary;
67
68 --PART-1
69 --Q1)SQL Query to fetch records that are present in one table but not in another table.
70 select E.* from EmployeeDetails E LEFT JOIN EmployeeSalary S on E.EmpId = S.EmpId where S.EmpId IS NULL;
71
72 --Q2)SQL query to fetch all the employees who are not working on any project.
73 select E.* from EmployeeDetails E LEFT JOIN EmployeeSalary S on E.EmpId = S.EmpId where S.Project IS NULL;
74
75 --Q3)SQL query to fetch all the Employees from EmployeeDetails who joined in the Year 2020.
76 select * from EmployeeDetails where year(DateOfJoining)=2020;
77
78 --Q4)Fetch all employees from EmployeeDetails who have a salary record in EmployeeSalary.
79 select E.EmpId,E.FullName from EmployeeDetails E inner join EmployeeSalary S on E.EmpId = S.EmpId;
80
81 --Q5)Write an SQL query to fetch a project-wise count of employees.
82 SELECT Project,COUNT(*) AS Employee_Count FROM EmployeeSalary GROUP BY Project;
83
84 --Q6)Fetch employee names and salaries even if the salary value is not present for the employee.
85 SELECT E.FullName,
86        S.Salary
87 FROM EmployeeDetails E
88 LEFT JOIN EmployeeSalary S
89 ON E.EmpId = S.EmpId;
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90
91 --Q7)Write an SQL query to fetch all the Employees who are also managers.
92 SELECT *
93 FROM EmployeeDetails
94 WHERE EmpId IN
95 (
96     SELECT DISTINCT ManagerId
97     FROM EmployeeDetails
98 );
99
100 --Q8)Write an SQL query to fetch duplicate records from EmployeeDetails.
101 SELECT EmpId,
102        FullName,
103        ManagerId,
104        DateOfJoining,
105        City,
106        COUNT(*) AS Duplicate_Count
107 FROM EmployeeDetails
108 GROUP BY EmpId, FullName, ManagerId, DateOfJoining, City
109 HAVING COUNT(*) > 1;
110
111 --Q9)Write an SQL query to fetch only odd rows from the table.
112 SELECT * FROM
113 (
114     SELECT *, ROW_NUMBER() OVER(ORDER BY EmpId) AS RowNum
115     FROM EmployeeDetails
116 ) X
117 WHERE RowNum % 2 = 1;
118
119 --Q10)Write a query to find the 3rd highest salary from a table without top or limit keyword.
120 SELECT DISTINCT Salary
121 FROM EmployeeSalary S1
122 WHERE 2 =
123 (
124     SELECT COUNT(DISTINCT Salary)
125     FROM EmployeeSalary S2
126     WHERE S2.Salary > S1.Salary
127 );
128
129 -----
130
131 --PART- 2
132 --Ques.1. Write an SQL query to fetch the EmpId and FullName of all the
133      employees working under the Manager with id - "986".
134 select EmpId,FullName from EmployeeDetails where ManagerId = '986';
135
136 --Ques.2. Write an SQL query to fetch the different projects available
137      from the EmployeeSalary table.
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135 select DISTINCT project from EmployeeSalary;
136
137 --Ques.3. Write an SQL query to fetch the count of employees working in
      project „P1“.
138 select count(*) as Emp_Count from EmployeeSalary where Project='P1';
139
140 --Ques.4. Write an SQL query to find the maximum, minimum, and average
      salary of the employees.
141 SELECT
142 MAX(Salary) AS Maximum_Salary,
143 MIN(Salary) AS Minimum_Salary,
144 AVG(Salary) AS Average_Salary
145 FROM EmployeeSalary;
146
147 --Ques.5. Write an SQL query to find the employee id whose salary lies in
      the range of 9000 and 15000.
148 select EmpId,Salary from EmployeeSalary where Salary between 9000 and
      15000;
149
150 --Ques.6. Write an SQL query to fetch those employees who live in Toronto
      and work under the manager with ManagerId - 321.
151 select * from EmployeeDetails where City = 'Toronto' and ManagerId =
      '321';
152
153 --Ques.7. Write an SQL query to fetch all the employees who either live in
      California or work under a manager with ManagerId - 321.
154 select * from EmployeeDetails where City = 'California' or ManagerId =
      '321';
155
156 --Ques.8. Write an SQL query to fetch all those employees who work on
      Projects other than P1.
157 Select * from EmployeeSalary where Project != 'P1';
158
159 --Ques.9. Write an SQL query to display the total salary of each employee
      adding the Salary with Variable value.
160 select EmpId,Salary,Variable,Salary+Variable as Total_Salary from
      EmployeeSalary;
161
162 --Ques.10. Write an SQL query to fetch the employees whose name begins
      with any two characters, followed by a text “hn” and ends with any
      sequence of characters.
163 SELECT * FROM EmployeeDetails WHERE FullName LIKE '__hn%';
164
165 -----
      -----
166 --PART - 3
167 --Ques.1 Write an SQL query to fetch all the EmpIds which are present in
      either of the tables - „EmployeeDetails“ and „EmployeeSalary“.
168 SELECT EmpId FROM EmployeeDetails UNION SELECT EmpId FROM
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EmployeeSalary;
169
170 --Ques.2 Write an SQL query to fetch common records between two tables.
171 SELECT E.EmpId,
172         E.FullName,
173         S.Project,
174         S.Salary
175 FROM EmployeeDetails E
176 INNER JOIN EmployeeSalary S
177 ON E.EmpId = S.EmpId;
178
179 --Ques.3. Write an SQL query to fetch records that are present in one      ↗
        table but not in another table.
180 SELECT E.* FROM EmployeeDetails E
181 LEFT JOIN EmployeeSalary S
182 ON E.EmpId = S.EmpId
183 WHERE S.EmpId IS NULL;
184
185 --Ques.4. Write an SQL query to fetch the EmpIds that are present in both  ↗
        the tables - „EmployeeDetails“ and „EmployeeSalary.
186 SELECT E.EmpId
187 FROM EmployeeDetails E
188 INNER JOIN EmployeeSalary S
189 ON E.EmpId = S.EmpId;
190
191 --Ques.5. Write an SQL query to fetch the EmpIds that are present in      ↗
        EmployeeDetails but not in EmployeeSalary.
192 SELECT E.EmpId
193 FROM EmployeeDetails E
194 LEFT JOIN EmployeeSalary S
195 ON E.EmpId = S.EmpId
196 WHERE S.EmpId IS NULL;
197
198 --Ques.6. Write an SQL query to fetch the employee`s full names and      ↗
        replace the space
199 SELECT FullName,
200        REPLACE(FullName, ' ', '_') AS Modified_Name
201 FROM EmployeeDetails;
202
203 --Ques.7. Write an SQL query to fetch the position of a given character(  ↗
        s) in a field.
204 SELECT FullName,
205        CHARINDEX('a', FullName) AS Position_Of_A
206 FROM EmployeeDetails;
207
208 --Ques.8. Write an SQL query to display both the EmpId and ManagerId      ↗
        together.
209 SELECT CONCAT(EmpId, ' - ', ManagerId)
210 AS Employee_Manager
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211 FROM EmployeeDetails;
212
213 --Ques.9. Write a query to fetch only the first name(string before space)  ↗
      from the FullName column of the EmployeeDetails table.
214 SELECT FullName,
215        LEFT(FullName,
216        CHARINDEX(' ',FullName)-1)
217        AS First_Name
218 FROM EmployeeDetails;
219
220 --Ques.10. Write an SQL query to uppercase the name of the employee and  ↗
      lowercase the city values.
221 SELECT
222 UPPER(FullName) AS Employee_Name,
223 LOWER(City) AS City
224 FROM EmployeeDetails;
225
226 -----  ↗
      -----
227 --PART - 4
228 --Ques.1. Write an SQL query to find the count of the total occurrences of  ↗
      a particular character - „n“ in the FullName field.
229 SELECT FullName, LEN(FullName) - LEN(REPLACE(LOWER(FullName), 'n', '')) AS  ↗
      Count_Of_N FROM EmployeeDetails;
230
231 --Ques.2. Write an SQL query to update the employee names by removing  ↗
      leading and trailing spaces.
232 UPDATE EmployeeDetails SET FullName = LTRIM(RTRIM(FullName));
233
234 --Ques.3. Fetch all the employees who are not working on any project.
235 SELECT E.*
236 FROM EmployeeDetails E
237 LEFT JOIN EmployeeSalary S
238 ON E.EmpId = S.EmpId
239 WHERE S.Project IS NULL;
240
241 --Ques.4. Write an SQL query to fetch employee names having a salary  ↗
      greater than or equal to 5000 and less than or equal to 10000.
242 select E.Fullname,S.Salary from EmployeeDetails E inner join  ↗
      EmployeeSalary S
243 on E.EmpId = S.EmpId where S.Salary BETWEEN 5000 AND 10000;
244
245 --Ques.5. Write an SQL query to find the current date-time.
246 select GETDATE() as Current_date_time;
247
248 --Ques.6. Write an SQL query to fetch all the Employee details from the  ↗
      EmployeeDetails table who joined in the Year 2020.
249 SELECT * FROM EmployeeDetails WHERE YEAR(DateOfJoining) = 2020;
250
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251 --Ques.7. Write an SQL query to fetch all employee records from the EmployeeDetails table who have a salary record in the EmployeeSalary table.
252 SELECT E.*
253 FROM EmployeeDetails E
254 INNER JOIN EmployeeSalary S
255 ON E.EmpId = S.EmpId;
256
257 --Ques.8. Write an SQL query to fetch the project-wise count of employees sorted by project's count in descending order.
258 SELECT Project,
259         COUNT(*) AS Employee_Count
260 FROM EmployeeSalary
261 GROUP BY Project
262 ORDER BY Employee_Count DESC;
263
264 --Ques.9. Write a query to fetch employee names and salary records. Display the employee details even if the salary record is not present for the employee.
265 SELECT E.FullName,
266         S.Salary
267 FROM EmployeeDetails E
268 LEFT JOIN EmployeeSalary S
269 ON E.EmpId = S.EmpId;
270
271 --Ques.10. Write an SQL query to join 3 tables.
272 CREATE TABLE Department(
273     DeptId INT,
274     EmpId INT,
275     DeptName VARCHAR(50)
276 );
277
278 SELECT E.EmpId,
279         E.FullName,
280         S.Project,
281         D.DeptName
282 FROM EmployeeDetails E
283 INNER JOIN EmployeeSalary S
284 ON E.EmpId = S.EmpId
285 INNER JOIN Department D
286 ON E.EmpId = D.EmpId;
287
```